When Banks Wait

Strategic Delay for Distressed Loan Resolution

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    - NYC's medallion resale market was described as "frozen" in 2016. (NYC Taxi Medallion Task Force '20)
    - By September 2017, Flushing Financial had restructured over 80% of its medallion loans through rate reductions and/or maturity extensions. (Flushing Financial '17)

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  - **Shipping following the global financial crisis**
    - Secondhand vessel markets became illiquid, while banks were reluctant to liquidate collateral backing troubled loans. (Drobetz, Haller, and Meier '16)

Introduction

- Why do delayed distressed loan resolution and low transaction activity occur together after a sectoral shock?

- This paper: waiting for information about collateral values can delay resolution because **price discovery is a public good**.
  - Resolving a loan generates information that other banks can use, giving them an incentive to wait for others to resolve first.
  - When banks wait, information production slows and uncertainty persists.
  - Use post-pandemic CRE distress as a laboratory to test this mechanism.

This Paper

1. Develop a dynamic model of delayed resolution based on **price discovery**
   - Resolution produces public information about common collateral values
   - Banks wait to learn from other banks' costly resolution decisions
   - Information production generates delays, without risk-shifting or information asymmetry

2. Construct **loan-level evidence** from the post-COVID CRE market
   - Link the mortgage origination record with property characteristics and bank regulatory data
   - Map bank-held CRE loans to subsequent extensions and resolutions

3. Test **three predictions** from the information mechanism
   1. More peer resolutions reduce subsequent loan extensions
   2. The effect is stronger when each resolution reveals more information
   3. The proportional effect is stronger under greater initial uncertainty

Literature

1. Zombie lending, forbearance, and delayed loss recognition
   Caballero et al. (’08), Bruche and Llobet (’14), Hu and Varas (’21), Becker and Ivashina (’21), Acharya et al. (’22)
   – Propose a public-information mechanism: banks wait to learn from other banks’ resolutions, slowing information production

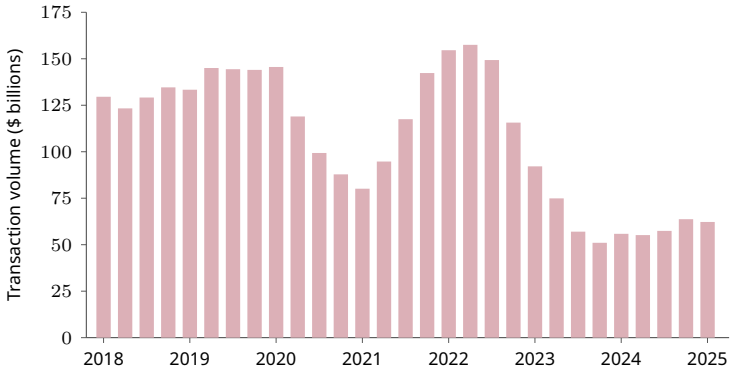
2. Post-COVID office shock and CRE lenders
   Gupta et al. (’22), Monte et al. (’23), Crosignani and Prazad (’24), Glancy et al. (’24), Anenberg et al. (’25)
   – Construct a novel loan-level dataset linking CRE loan histories to property transactions to study resolution decisions, including those of regional and small banks

3. Social learning, information production, and slow recovery
   Chamley and Gale (’94), Caplin and Leahy (’94), Veldkamp (’05), Gorton and Ordoñez (’14), Fajgelbaum (’17)
   – Develop a dynamic model of distressed-loan resolution with endogenous public learning and test its predictions using loan-level data

2. Motivating Stylized Facts

Office Transaction Volume Fell after 2022

Figure: U.S. Office Transaction Volume

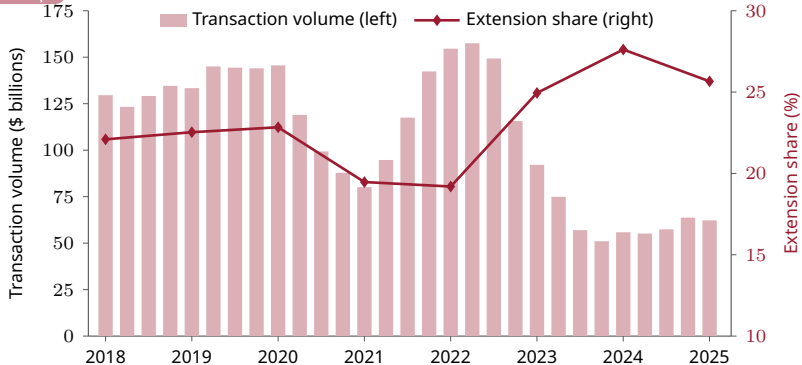


Source: RCA.

Loan Extensions Became More Common after 2022

- Extension increases appear across bank capital buffers (Glancy, 2026) and bank types.

▸ Bank Decomp.



Notes: Annual extension share: ext./ (refi. + ext. + default), pooled across banks. 2025 is partial.

Sources: RCA (quarterly volume), CoreLogic and FFIEC.

Link the Low Activity with Bank Extensions

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- A follow-up question asked how office prices were changing and when they would stabilize. Thomas replied:

> *"You're asking a very good but unanswerable question, and I'll explain why. We tried.*
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> *I try every quarter to give all of you comparable market deals. So we all understand together what pricing is. And this quarter, I could not come up with one."*

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→ I'll propose a channel where **delayed resolution can arise from price discovery**

3. Model

Model Setup

- Time is discrete. Two types of agents: $\bar{N}$ banks and $M(> \bar{N})$ developers
- Each bank holds one distressed loan backed by one indivisible unit of collateral. To hold the asset, in each period, each bank will pay a **privately drawn holding cost**

$$c_{i,t} \overset{\text{iid}}{\sim} G.$$

- The assets have an underlying value that can only be realized by the developers:

$$\underbrace{\theta}_{\text{common value}} + \underbrace{\varepsilon_i}_{\text{idiosyncratic value}}, \quad \varepsilon_i \overset{\text{iid}}{\sim} N(0, \delta^{-1})$$

- The component θ is fixed but unknown, with a common belief across agents at $t = 0$

$$\theta \mid \mathcal{I}_0 \sim N\left(\mu_0, \gamma_0^{-1}\right).$$

All the information about θ is public; there is no information asymmetry Common value uncertainty

Information Comes from Loan Resolutions

- In each period, each bank decides whether to sell the asset to a developer (**resolve**) or keep the asset (**wait**)
- Resolution is the only source of new public information: the asset value is realized after the transaction, generating a public signal

$$x_i = \theta + \varepsilon_i, \qquad \varepsilon_i \overset{\text{iid}}{\sim} N(0, \delta^{-1}).$$

  When n banks resolve, the combined signal has precision $n\delta$.

- State variables: public belief mean μ_t, public belief precision γ_t, and number of remaining unresolved banks N_t.



Public Uncertainty Lowers the Resolution Price

- Developers are short-lived and competitive, with mean-variance preferences.
- Denote the price of a distressed asset by P_t. Developers maximize

$$\max_k \ \mathbb{E}_t\left[\sum_{i=1}^{k}(\theta+\varepsilon_i) - kP_t\right] - \frac{a}{2}\operatorname{Var}_t\left[\sum_{i=1}^{k}(\theta+\varepsilon_i)\right]$$
$$= k\left(\mu_t - P_t - \frac{a}{2}\frac{1}{\delta}\right) - \frac{a}{2}k^2\frac{1}{\gamma_t}.$$

- In equilibrium, developers earn zero profits, and each developer holds at most one unit of the asset, which leads to the price function

$$P_t = \mu_t - \frac{a}{2\gamma_t} - \frac{a}{2\delta}.$$

Low public precision γ_t produces a larger uncertainty discount and a lower current resolution payoff.

Banks Decide When to Resolve the Distressed Loans

- Banks are risk neutral with no discounting and choose between resolving and waiting in each period:

$$V(\mu_t, \gamma_t, N_t, c_{i,t}) = \max \left\{ \underbrace{V^R(\mu_t, \gamma_t)}_{\text{resolution value}}, \underbrace{V^W(\mu_t, \gamma_t, N_t, c_{i,t})}_{\text{waiting value}} \right\}.$$

where

$$V^R(\mu_t, \gamma_t) = P(\mu_t, \gamma_t) = \mu_t - \frac{a}{2\gamma_t} - \frac{a}{2\delta},$$

$$V^W(\mu_t, \gamma_t, N_t, c_{i,t}) = \underbrace{-c_{i,t}}_{\text{idiosyncratic holding cost}} + \mathbb{E}_t \left[V(\mu_{t+1}, \gamma_{t+1}, N_{t+1}, c_{i,t+1}) \right].$$

- Define the ex ante value for a bank before drawing c as $\mathcal{V}$. It is affine in the belief mean:

$$\mathcal{V}(\mu_t, \gamma_t, N_t) \equiv \int V(\mu_t, \gamma_t, N_t, c_{i,t})\, dG(c) = \mu_t + v(\gamma_t, N_t).$$

→ **μ_t does not enter the policy function; the relevant state variables are γ_t and N_t.**

Symmetric Cutoff Strategy for Banks

- In equilibrium, banks follow a symmetric cutoff strategy:

$$\text{resolve if and only if } c \geq c^*(\gamma, N),$$

  so we can denote the resolution probability for each bank as

$$q^*(\gamma, N) = 1 - G(c^*(\gamma, N))$$

- Then the number of resolutions n_t follows a Binomial distribution with probability q^*:

$$n_t \mid (\gamma_t, N_t) \sim \text{Binomial}\big(N_t, q^*(\gamma_t, N_t)\big),$$

  and public precision and the number of unresolved banks evolve as

$$(\gamma_{t+1}, N_{t+1}) = (\gamma_t + n_t\delta, N_t - n_t).$$

- We can derive the optimal cutoff function and value function v from a fixed point problem:

$$c^*(\gamma, N) = \frac{a}{2\gamma} + \frac{a}{2\delta} + \mathbb{E}_n[v(\gamma + n\delta, N - n)], \quad n \sim \text{Binomial}(N - 1, q^*(\gamma, N)).$$

Intuition: Information Free-Riding Creates Delay

- After an aggregate shock in a market, collateral values fall and uncertainty rises for similar collateral in the affected market.

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- **Why delay happens**: the resolving bank does not internalize the information benefit
  - Within-period substitution: when other banks are more likely to resolve, I prefer to wait.
  - Information free-riding $\rightarrow$ each bank wants others to resolve first.

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  - Within-period substitution: when other banks are more likely to resolve, I prefer to wait.
  - Information free-riding $\rightarrow$ each bank wants others to resolve first.
- **Why delay can be sustained**:
  - Across-period complementarity: resolution improves information and makes future resolution more attractive.
  - If the market starts with high uncertainty, delay can persist for longer and the market can remain cold.

Proposition 1: One Additional Realized Resolution Makes Future Resolutions More Likely

- Starting from the same state (γ_t, N_t), compare histories with n and $n+1$ realized resolutions.
- **One additional realized resolution increases the resolution probability of a remaining bank next period.**

Proposition

For $n = 0, \ldots, N_t - 2$, define

$$q_n^{t+1}(\gamma_t, N_t; \delta) \equiv q^*(\gamma_t + n\delta, N_t - n).$$

Then

$$\Delta_n q_{t+1}(\gamma_t, N_t; \delta) \equiv q_{n+1}^{t+1}(\gamma_t, N_t; \delta) - q_n^{t+1}(\gamma_t, N_t; \delta) \geq 0.$$

$\Delta_n q_{t+1}$ measures the change in next-period resolution probability associated with one additional current resolution.

Why Resolutions Are Complementary Across Periods

$\Delta_n q_{t+1}$ can be decomposed into

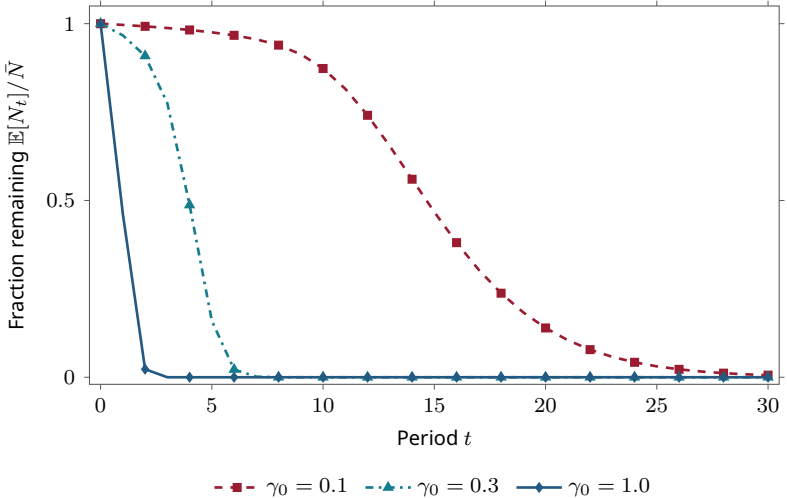
$$\Delta_n q_{t+1}(\gamma_t, N_t; \delta) = \underbrace{q^*(\gamma_t + (n+1)\delta, N_t - (n+1)) - q^*(\gamma_t + n\delta, N_t - (n+1))}_{\text{greater public precision} \geq 0}$$
$$+ \underbrace{q^*(\gamma_t + n\delta, N_t - (n+1)) - q^*(\gamma_t + n\delta, N_t - n)}_{\text{fewer unresolved peers} \geq 0}$$

Intuition:

Lower resolution in t
$\rightarrow$ lower γ_{t+1} and higher N_{t+1}
$\rightarrow$ **lower price tomorrow and more opportunities to learn from peers**
$\rightarrow$ lower resolution in $t+1$

so high uncertainty can persist for a long time if we start from low γ_0.

Higher Initial Uncertainty Sustains Resolution Delay



Proposition 2: High Signal Precision Strengthens Complementarity

- Starting from the same initial state (γ_t, N_t), compare histories with n and $n+1$ realized resolutions under two signal precisions.
- Under the conditions below, signal precision $\uparrow \rightarrow$ resolution-probability response to an additional resolution $\uparrow$.

Proposition

For $\delta^H > \delta^L > 0$, there exists a finite a^* such that, when $a \geq a^*$ and γ_t is sufficiently low,

$$\Delta_n q_{t+1}(\gamma_t, N_t; \delta^H) > \Delta_n q_{t+1}(\gamma_t, N_t; \delta^L), \qquad n = 0, \ldots, N_t - 3.$$

Intuition: the uncertainty discount change from one signal realization,

$$\frac{a}{2}\left(\frac{1}{\gamma_t} - \frac{1}{\gamma_t + \boldsymbol{\delta}}\right),$$

increases with δ: $\delta \uparrow \rightarrow$ price change next period $\uparrow \rightarrow$ resolution-prob. response $\uparrow$.

Proposition 3: Initial Uncertainty Amplifies the Proportional Response

- Starting from the same (N_t), but with different γ_t, compare the **proportional response** of the next period's resolution probability to an additional realized resolution
- Public precision $\gamma_t \downarrow \rightarrow$ proportional change in resolution prob. from an additional resolution $\uparrow$.

Proposition

For any $0 < \gamma^L < \gamma^H$, there exists a finite a^* such that, when $a \geq a^*, N_t \geq 3$,

$$\Delta_n \log q_{t+1}(\gamma^L, N_t; \delta) > \Delta_n \log q_{t+1}(\gamma^H, N_t; \delta),$$

where

$$\Delta_n \log q_{t+1}(\gamma_t, N_t; \delta) \equiv \log q^*(\gamma_t + (n+1)\delta, N_t - n - 1) - \log q^*(\gamma_t + n\delta, N_t - n).$$

Intuition: the uncertainty discount change from one signal realization,

$$\frac{a}{2}\left(\frac{1}{\gamma_t} - \frac{1}{\gamma_t + \delta}\right),$$

decreases with γ_t: $\gamma_t \downarrow \rightarrow$ price change next period $\uparrow \rightarrow$ proportional resolution-prob. response $\uparrow$.

Testable Predictions from the Model

1. **Across-period complementarity:** early peer resolutions reduce later extension among loans that remain unresolved.
   - Loan-level regressions and shift-share IV.

2. **Resolution Informativeness:** the reduction in extension is larger when realized resolutions reveal more public information.
   - Variation in public disclosure of transaction prices.

3. **Initial valuation uncertainty:** the proportional increase in resolution probability is larger when initial public precision is lower.
   - Predetermined WFH exposure as a proxy for uncertainty.

4. Empirical Tests

Data

- **CRE Loan Records: CoreLogic Mortgage Data**
  - County land records of mortgage origination, modifications, and refinancing
  - Include: Loan amounts, origination and maturity dates, lender names, and property ID
  - Link the non-residential-non-farm loan records over time to construct a loan-quarter panel

- **Property Information and Valuation**
  - CoreLogic tax and ownership records: property characteristics, assessed values, sales, and foreclosure-related transfers
  - CompStak: commercial lease records; RCA CPPI: commercial property price indices

- **Bank Characteristics: Call Reports and FR Y-9C**
  - Match mortgage lenders to regulatory identifiers and their ultimate parent institutions
  - Link to quarterly bank balance sheets, capital, and CRE exposure

The cleaned panel contains 0.63 million loans and covers 2,089 counties and 2.9k ultimate parent banks

Sample and Timing

- I follow CRE loans outstanding at the beginning of 2020 through 2025Q2.
- I record the events for all loans, including extensions, refinancing, defaults, and repayments, and link each loan to property transaction records.
- I divide the sample into two periods:
  - **Earlier period (t_1):** 2020Q1–2021Q4.
  - **Later period (t_2):** 2022Q1–2025Q2.
- The tests focus on the correlation between resolution events in t_1 and loan extensions in t_2.
- Mapping to the model:
  - Extensions show banks choose to wait
  - Resolution events provide observable outcomes for the collateral, which may help other lenders assess similar properties

Constructing the Regression Variables

- **First, record whether each loan is extended.**
  For each loan still outstanding at the start of 2022:
  - Set $Ext_i = 1$ if its maturity is extended during t_2 through modification or same-lender refinancing.

- **Then, count earlier sales and defaults in its market.**
  During t_1, in the same county (c) and property type (p), count:
  - Sale payoff: the property is sold and the loan is repaid in the same quarter.
  - Default: the property transfers from the borrower to the creditor.
  - Exclude other refinancing and repayments without a recorded property sale.

  Excluding loans held by the same parent bank ($-b$), define the **resolution count**:

$$ResCounts_{c,p,-b,t_1} = SaleCounts_{c,p,-b,t_1} + DefaultCounts_{c,p,-b,t_1}.$$

Prediction 1: Empirical Specification

More earlier resolutions in a market predict a lower probability of subsequent loan extension: $\beta < 0$.

$$Ext_{i,c,p,b,t_2} = \alpha + \boldsymbol{\beta}\, ResCounts_{c,p,-b,t_1} + \Theta' W_{i,c,p,b,t_0} + \lambda_c + \kappa_p + \psi_b + \varepsilon_{i,c,p,b,t_2}.$$

- i: loan; c: county; p: property type; b: parent bank.
- $ResCounts$: earlier sale payoffs and defaults at other banks
- W: controls including loan characteristics, initial market size, WFH exposure, etc.
  - WFH exposure: a measure of exposure to working from home, based on pre-pandemic local industry composition and industry level WFH feasibility (Dingel and Neiman, 2020).
- **Endogeneity concern:** unobserved local shocks may affect both earlier resolutions and later extensions.
  - Next: a shift-share IV to address this concern.

Prediction 1: Shift-Share IV Design

- Model motivation: idiosyncratic waiting costs generate variation in resolution decisions.
- Identifying variation: bank-level resolution pressure originating outside the local market.
- **Intuition**: banks facing greater resolution pressure may resolve more loans across their markets.
  - Markets where these banks initially held more loans are more exposed.
- **Construction**: Following Greenstone, Mas, and Nguyen (2020).
  - Shifter: estimated bank-level resolution shock, separating bank and market effects in resolution rates outside the current market.
  - Share: initial local loan shares of peer banks, excluding the loan's own parent bank.
  - Rate instrument: share-weighted average of estimated bank shocks.
  - Count instrument: rate instrument multiplied by the initial peer-loan stock.

Jointly instrument for informative-resolution rates and counts.

Prediction 1: $\beta < 0$, peer resolutions predict less subsequent extension

	(1) Baseline	(2) + Bank FE	(3) + Local controls	(4) Joint IV
Resolution counts (β)	**-0.278****	**-0.272*****	**-0.264*****	**-0.229*****
	(0.138)	(0.077)	(0.076)	(0.066)
WFH exposure	0.257*	0.314**	0.349**	0.505***
	(0.144)	(0.137)	(0.141)	(0.178)
Information-event rate (pp)			-0.025**	-0.306
			(0.012)	(0.192)
Local Controls	N	N	Y	Y
Bank FE	N	Y	Y	Y
County FE, Property Type FE	Y	Y	Y	Y
Market Size and Loan Level Controls	Y	Y	Y	Y
Observations	626,839	626,839	626,839	626,839
R-squared	0.0451	0.0729	0.0729	
KP rk Wald F				8.12
SW F: ResCounts				25.14
SW F: InfRate				16.25

Notes: Standard errors clustered by county. Control definitions

Prediction 2: Resolution Informativeness

When each resolution is more informative, an additional resolution will lead to a larger reduction in subsequent extension.

- **State-level transaction-price disclosure policies**
  - Disclosure states: sale prices become public records.
  - Non-disclosure states: public disclosure of sale prices is not required.

- Sale payoff: disclosed prices provide additional information about property values.
  $\rightarrow$ Larger reduction in extension in disclosure states.

- Defaults: borrower-to-creditor ownership transfers do not themselves reveal a market sale price.
  $\rightarrow$ the disclosure policy doesn't change the information content for defaults

Prediction 2: Empirical Specification

Let $D_c = 1$ for counties in states requiring public disclosure of sale prices, and zero otherwise.

$$Ext_{i,c,p,b,t_2} = \alpha + \beta_S \, SaleCounts_{c,p,-b,t_1} + \boldsymbol{\theta_S} \, (SaleCounts_{c,p,-b,t_1} \times D_c) + \Theta' W_{i,c,p,b,t_0} + \lambda_c + \kappa_p + \psi_b + \varepsilon_{i,c,p,b,t_2}.$$

- **Without required price disclosure ($D_c = 0$):**
  β_S measures the relationship between earlier sale payoffs and subsequent extensions.
- **With required price disclosure ($D_c = 1$):**
  The corresponding relationship is $\beta_S + \theta_S$.
- **Prediction: $\theta_S < 0$.**
  Earlier sale payoffs predict a larger reduction in subsequent extensions where transaction prices are publicly disclosed.

Joint specification

Prediction 2: $\theta_S < 0$, Price disclosure strengthens the sale-payoff effect

	(1) Baseline	(2) + Bank FE	(3) + Local controls
Sale payoffs $\times$ Disclosure (θ_S)	**-0.424****	**-0.389****	**-0.418****
	(0.205)	(0.184)	(0.188)
Sale payoffs	-0.586***	-0.569***	-0.493***
	(0.154)	(0.136)	(0.148)
WFH exposure	0.315**	0.351***	0.405***
	(0.138)	(0.132)	(0.140)
Local controls	N	N	Y
Bank FE	N	Y	Y
County FE, Property Type FE	Y	Y	Y
Market Size and Loan Level Controls	Y	Y	Y
Observations	626,839	626,839	626,839
R-squared	0.0451	0.0728	0.0728

Notes: Standard errors clustered by county. Control definitions

Prediction 3: Initial Uncertainty

When the market starts from greater initial uncertainty, one additional resolution leads to a higher proportional increase in resolution probability.

- Intuition: when banks are less certain about property values, information from peer resolutions matters more.
- **Design**: use predetermined WFH exposure as a proxy for initial valuation uncertainty.
- WFH changes property demand; its impact on property values remains uncertain.
- More WFH-exposed markets
  $\rightarrow$ Greater uncertainty about future property values.

Prediction 3: Empirical Specification

Prediction 3 concerns proportional changes in resolution probability.

- $Resolution_i = 1$ if loan i experiences a default, sale payoff, or other repayment during t_2
- Estimate an exponential conditional mean using Poisson pseudo-maximum likelihood (PPML).

$$\mathbb{E}[Resolution_{i,c,p,b,t_2} \mid \mathcal{X}_i] = \exp\left\{\begin{array}{l}\alpha + \zeta_H \left(ResCounts_{c,p,-b,t_1} \times WFH_{c,p}\right) \\ + \zeta_X ResCounts_{c,p,-b,t_1} + \rho WFH_{c,p} \\ + \Theta' W_{i,c,p,b,t_0} + \lambda_c + \kappa_p + \psi_b\end{array}\right\}.$$

- **Prediction:** $\zeta_H > 0$

  Higher WFH exposure amplifies the proportional increase in resolution probability following additional peer resolutions.

WFH exposure measure

Prediction 3: $\zeta_H > 0$, WFH exposure amplifies the proportional resolution response

	(1) Baseline	(2) + Bank FE	(3) + Local controls
Resolution counts × WFH (ζ_H)	**0.124*****	**0.123*****	**0.128*****
	(0.030)	(0.030)	(0.030)
Resolution counts	0.202***	0.220***	0.284***
	(0.057)	(0.056)	(0.062)
WFH exposure	-0.016	-0.014	-0.013
	(0.070)	(0.063)	(0.062)
Estimator	PPML	PPML	PPML
Local controls	N	N	Y
Bank FE	N	Y	Y
County FE, Property Type FE	Y	Y	Y
Market Size and Loan Level Controls	Y	Y	Y
Observations	618,945	618,945	618,945

Notes: PPML estimates; outcome is the resolution indicator. Standard errors clustered by county.

Control definitions

Additional Tests

Additional design for mechanism support

- Good versus bad price news
  - Does extension decline after both good and bad news, or primarily after bad news that pressures banks to recognize losses?

Robustness Tests

- Prediction 2: Price disclosure for sale payoffs versus defaults
  - Examine whether disclosure also changes the relationship between earlier defaults and subsequent extensions.
- Prediction 3: Initial uncertainty and loan extensions
  - Examine whether greater initial uncertainty strengthens the decline in extension probability following earlier resolutions.
- Robustness test on window cutoff

5. Conclusion

Delayed Resolution Can Reflect an Information Problem

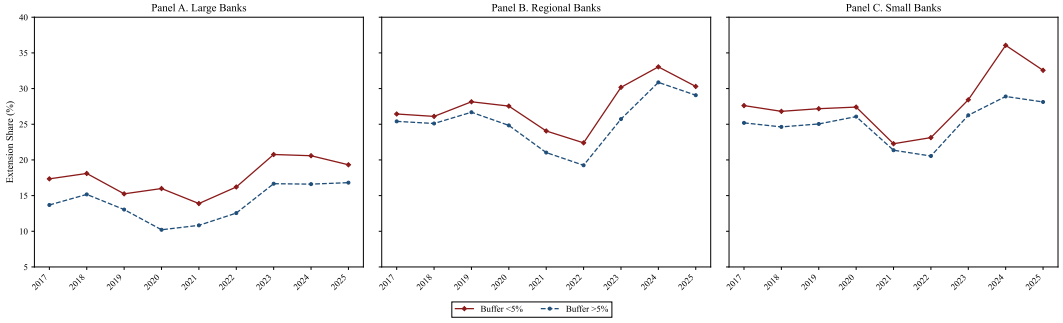
- **Question:** Why do delayed loan resolution and low transaction activity occur together after a sectoral shock?
- **Model:** Resolving a loan produces information about collateral values, giving other banks an incentive to wait and learn.
- **Empirical evidence:**
  - More earlier resolutions predict fewer subsequent extensions, with supporting evidence from shift-share IV estimates.
  - This relationship is stronger with public price disclosure.
  - Greater WFH exposure predicts a larger proportional resolution response.
  - Both good- and bad-news transactions predict fewer extensions.

Individually optimal waiting can prolong aggregate distress because price discovery is a public good.

Appendix

Extensions Rose Across Bank Capital Buffers

- FR Y-14Q data (Glancy, 2026): extensions at banks comfortably above their capital requirements
- Land records: increase in extension share across bank types and banks' capital health after 2022

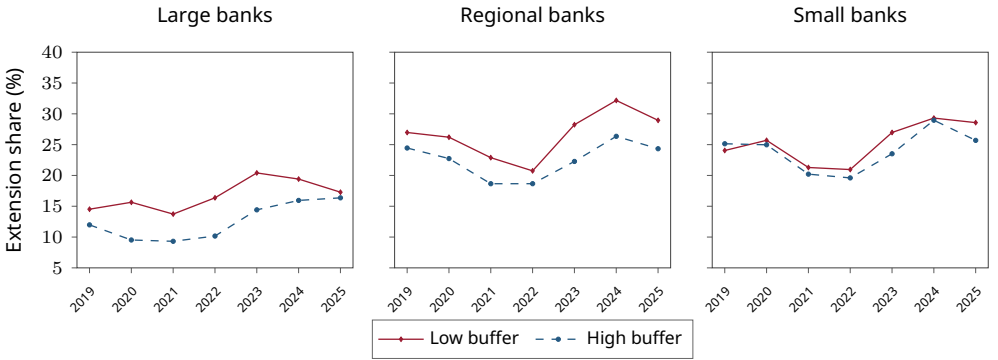


Notes: Buffer: 2019Q4 MtM CET1. Extension share: ext./ (refi. + ext. + default). Sources: CoreLogic and FFIEC.

Asset cutoffs: Federal Reserve (2019), $100B; FFIEC CRA (2019), $1.284B. ▸ Robustness ▸ Large banks ◂ Return to motivation

Extension Share with a Different Cutoff: 2019Q4 Median Buffer

- Split banks by the peer-median MtM CET1 buffer within each bank-size group: Large, 4.29%; Regional, 6.37%; Small, 8.65%.



Notes: Buffer: 2019Q4 MtM CET1. Peer medians use CRE-active mortgage-parent banks with positive buffers. Extension share: ext./ (refi. + ext. + default). Sources: CoreLogic and FFIEC.

Return to motivation

Waiting costs: interpretation and role

- The transitory cost $c_{i,t}$ summarizes bank-specific reasons for resolving at different times: servicing and monitoring expenses, liquidity needs, capital pressure, and regulatory intervention.
- Independence from θ isolates learning from realized resolution outcomes. Informative waiting costs would also generate learning from banks' resolution decisions, requiring a richer public belief state.
- The cost shock **purifies the equilibrium**: banks play pure cutoff strategies where a complete-information version would require mixing. The continuous distribution yields a smooth resolution probability.
- If every bank instead faces $c_{i,t} \equiv \kappa > 0$, an interior symmetric equilibrium generally requires randomization between resolving and waiting. The fixed-cost benchmark retains strategic substitution and across-period complementarity.

Return to environment

Interpreting common valuation uncertainty

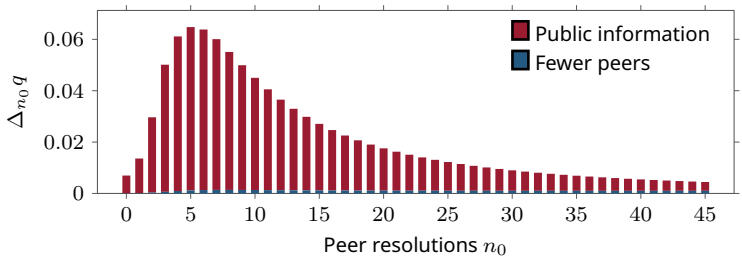
- θ is the component of asset values shared across banks; the underlying assets need not be identical.
- In CRE, properties in the same market face related demand, financing, and redevelopment conditions. Large demand shifts create uncertainty about post-shock values and alternative uses.
- Resolution reveals a noisy signal about this common component, making one bank's information useful to others. Entirely idiosyncratic asset values would eliminate this informational spillover.
- δ captures how observable and informative an event is. If outcomes were observed only privately, the reduced public-belief state used here would no longer suffice.

Return to environment

Decomposition: Precision Channel Dominates the Complementarity

Decomposition of the across-period complementarity. Let $N = N_- - m$ and $\gamma = \gamma_- + m\delta$. Then the realized resolution's impact on q given m can be decomposed into

$$\Delta_m q = \underbrace{q(N-1,\gamma) - q(N,\gamma)}_{\text{fewer unresolved peers}} + \underbrace{q(N-1,\gamma+\delta) - q(N-1,\gamma)}_{\text{greater public precision}}.$$



Return to complementarity

Equivalent Definition: Recursive Equilibrium

To record the transition generating the current state, define

$$\widehat{S} = (\mu_-, \gamma_-, N_-, m, X),$$

where m resolutions occurred at the preceding state and X is their average signal. The current state is recovered from

$$S = (\mu, \gamma, N) = \left(\frac{\gamma_- \mu_- + m\delta X}{\gamma_- + m\delta},\ \gamma_- + m\delta,\ N_- - m \right),$$

with $\mu = \mu_-$ when $m = 0$.
An equivalent **symmetric recursive equilibrium** $\{s^*, q, V, \mathcal{V}, P\}$ satisfies:

1. **Bellman optimality**

$$V(S, c) = \max\left\{ P(S),\ -c + \mathbb{E}_{n, X'}\left[\mathcal{V}(S'(n)) \mid S \right] \right\}, \qquad \mathcal{V}(S) = \int V(S, c)\, dG(c),$$

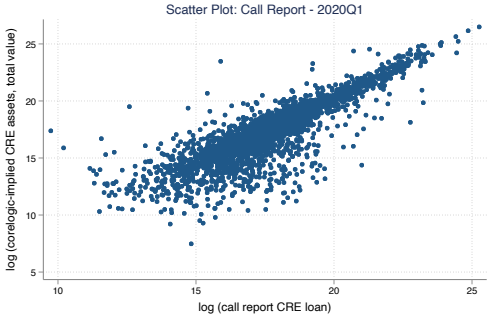
   where $n \sim \text{Binomial}(N - 1, q(S))$

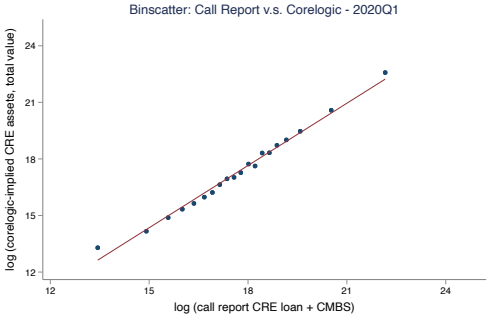
2. **Strategy consistency** $s^*(S, c) = \mathbf{1}\{c \geq c^*(S)\}, \qquad q(S) = 1 - G(c^*(S)).$
3. **Resolution price** $P(S) = \mu - \frac{a}{2\gamma} - \frac{a}{2\delta}$
4. **Aggregate law of motion** $m' \sim \text{Binomial}(N, q(S)), \qquad \widehat{S}' = (\mu, \gamma, N, m', X')$

Return to equilibrium

CoreLogic Coverage is Representative Across Bank CRE Holdings

Following Anenberg et al. (2025), compare banks' CRE holdings—measured as the sum of loan origination amounts—with their CRE balances reported in regulatory data

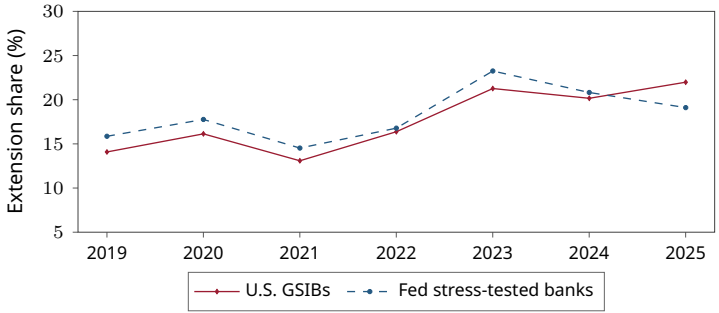




Return to data

Extension Trends among GSIBs and Stress-Tested Banks

- Pooled extension shares for eight U.S. GSIBs and 27 banks in the stress-tested sample. The broader group includes all eight GSIBs.



Notes: Extension share: ext./ (refi. + ext. + default). Fixed 2019 parent attribution; GSIB status at 2019Q4. The Fed's 2024 capital-requirements list has 32 banks: 30 matched, 27 represented in the mortgage sample.

Sources: CoreLogic and FFIEC. 2025 is partial. Return to motivation

Introduce a Measure of Predetermined WFH Exposure

- WFH exposure may affect both earlier peer resolutions and later loan extensions; control for this common local shock.
- A property market is more exposed when **its pre-pandemic lease demand comes from industries whose jobs can be done from home**
- Start with county (c)-industry (j) employment shares, $s_{c,j}$ (County Business Patterns, 2019).
- Map industries (j) to property types (p) using each industry's lease-value shares, $u_{j,p}$ (CompStak, 2009–2019).
- Use these weights to average the industry (j)-level WFH feasibility, h_j (Dingel and Neiman, 2020):

$$H_{c,p} = \frac{\sum_j s_{c,j}\, u_{j,p}\, h_j}{\sum_j s_{c,j}\, u_{j,p}}.$$

  $\rightarrow$ measure how likely the county–property-type market is to be affected by the WFH shock

- Pre-pandemic shares and job-based feasibility give a predetermined control, avoiding a potential bad-control problem

Shift-Share IV: Construction

Let $m = (c, p)$ denote a market, b the loan's parent bank, and k a peer bank.

- **Estimate bank-level resolution shocks**. Excluding market m, estimate across markets $\ell \neq m$:

$$\frac{ResCounts_{\ell,k,t_1}}{N_{\ell,k,t_0}} = d_\ell + Shock_k + e_{\ell,k}.$$

  Market fixed effects d_ℓ absorb common local conditions; denote the estimated bank effect by $\widehat{Shock}_k^{-m}$.

- **Construct initial peer-bank shares**:

$$Share_{m,k,t_0}^{-b} = \frac{N_{m,k,t_0}}{N_{m,-b,t_0}}, \qquad N_{m,-b,t_0} = \sum_{k \neq b} N_{m,k,t_0}.$$

- **Construct rate and count instruments**:

$$Z_{m,-b}^R = \sum_{k \neq b} Share_{m,k,t_0}^{-b} \widehat{Shock}_k^{-m}, \qquad Z_{m,-b}^X = N_{m,-b,t_0} Z_{m,-b}^R.$$

Following Greenstone, Mas, and Nguyen (2020). Before rescaling, $X = ResCounts$ and $R = X/N$. Both instruments enter both first-stage equations.

Shift-Share IV: First-Stage Results

Relevance: the two instruments must generate independent variation in peer resolution rates and counts, conditional on controls.

	Rate R	**Count** X
Rate instrument	0.444^{***}	-0.005
	(0.110)	(0.004)
Count instrument	-0.032	0.388^{***}
	(0.021)	(0.104)
Conditional SW F	16.25	25.14
Joint diagnostics	**Statistic**	p**-value**
KP rk LM	11.65	0.0006
KP rk Wald F	8.12	–
Anderson–Rubin F	4.06	0.0174

- Both instruments enter both first stages; KP rk Wald $F = 8.12$ warrants caution about joint strength.
- Anderson–Rubin rejects the joint null $\beta_R = \beta_X = 0$ at 5% ($p = 0.0174$).

Shift-Share IV: Exclusion Restriction

Conditional on controls and fixed effects, exposure to peer banks' outside-market resolution shocks affects later extension only through earlier peer resolution rates and counts.

- **Leave-out design limits mechanical feedback**.
  - Exclude the current market when estimating bank shocks.
  - Exclude the loan's own parent bank when aggregating shocks.
- **Concern 1: reduced refinancing supply**.
  - Peer-bank pressure may reduce outside refinancing options.
  - If this induces existing lenders to extend, the channel predicts more extension, opposite to $\beta_X < 0$.
- **Concern 2: bank sorting across markets**.
  - Control for predetermined WFH exposure, liquidity, competition, and fixed effects.
  - Proposed robustness check: reconstruct exposure using 2015 bank shares.

Back

Empirical Controls

- **Loan-level controls**
  - Log loan age.
  - Loan-to-value ratio at origination.
  - Log time to maturity.
- **Local controls**
  - Property liquidity: pre-pandemic annual turnover and log transaction counts, using 2015–2019 data.
  - Bank competition: HHI based on bank loan amounts.
  - Market size: 20 bins of initial peer-loan stock.
- **Additional covariates**
  - Predetermined WFH exposure.
  - Peer resolution rate (Prediction 1, columns 3–4): informative-resolution count divided by initial peer-loan stock.

◂ Prediction 1 ◂ Prediction 2 ◂ Prediction 3: Resolution ◂ Prediction 3: Extension

Property Transaction-Price Disclosure

- **The distinction concerns public access to sale prices**.
  - Most counties require recording of property ownership transfers; however, recording an ownership transfer does not necessarily make its transaction price public.
  - Disclosure states provide public access to sale-price information through property or transfer records.
- **Non-disclosure can take different forms**.
  - e.g. Texas: sale prices are not required to be reported to appraisal districts.
  - e.g. Montana: transfer certificates are collected but generally confidential, subject to statutory exceptions.
- **Interpretation for the empirical test**.
  - Public sale prices make valuation signals available beyond the transaction participants.
  - Interpret disclosure as increasing the public informativeness of sale payoffs.

Sources: Texas Comptroller (2015); Montana Code, Sec. 15-7-308; Wisconsin Department of Revenue.

Prediction 3: Implications for Loan Extension

Link initial uncertainty more directly to banks' extension decisions.

$\rightarrow$ Under a stronger condition, the model predicts the response of extension probability to peer resolutions in levels:

> When initial uncertainty is sufficiently high, greater uncertainty amplifies the reduction in extension probability following an additional peer resolution.

- $Ext_i = 1$ if loan i is extended during t_2; zero otherwise

$$\begin{aligned} Ext_{i,c,p,b,t_2} = \alpha &+ \beta_H \left(ResCounts_{c,p,-b,t_1} \times WFH_{c,p} \right) \\ &+ \beta_X ResCounts_{c,p,-b,t_1} + \rho WFH_{c,p} \\ &+ \Theta' W_{i,c,p,b,t_0} + \lambda_c + \kappa_p + \psi_b + \varepsilon_{i,c,p,b,t_2}. \end{aligned}$$

- **Prediction under sufficiently high uncertainty:** $\beta_H < 0$
  Higher WFH exposure amplifies the reduction in extension probability.

Prediction 3: Extension Results

	(1) Baseline	(2) + Bank FE	(3) + Local controls
Resolution counts × WFH (β_H)	**-0.190*****	**-0.099***	**-0.099***
	(0.063)	(0.051)	(0.054)
Resolution counts (centered)	-0.514***	-0.395***	-0.389***
	(0.150)	(0.112)	(0.119)
WFH exposure	0.343**	0.358***	0.394***
	(0.145)	(0.138)	(0.141)
Estimator	OLS	OLS	OLS
Local controls	N	N	Y
Bank FE	N	Y	Y
County FE, Property Type FE	Y	Y	Y
Market Size and Loan Level Controls	Y	Y	Y
Observations	626,839	626,839	626,839
R-squared	0.0452	0.0729	0.0729

Notes: OLS estimates; outcome is the extension indicator. Standard errors clustered by county.

Control definitions

Alternative Mechanism

- Alternative mechanism 1: adverse information forces loss recognition
  - Follow the logic of Gorton and Ordoñez (2014), *Collateral Crises*.
  - Low peer sale prices make continued forbearance harder to justify.
  - Prediction: extension falls primarily following bad news.
- Proposed mechanism: information reduces uncertainty
  - In the model, the resolution cutoff depends on belief precision, not the belief mean μ.
  - Both good and bad news can reduce valuation uncertainty.
  - Prediction: extension can fall following either type of news.

Additional Tests

Good versus Bad News: Empirical Specification

Focus on price-disclosure states, where each sale-payoff transaction has an observed price

- Calculate the "price signals" as the **difference between transaction price and the most recent assessed value**.
- Split $SaleCounts$ at agg. mean of "price signals": relatively good news (G) and bad news (B).
- Aggregate each group into event counts within the market, excluding the loan's own parent bank.

$$Ext_{i,c,p,b,t_2} = \alpha + \boldsymbol{\beta_G}\ SaleCounts^{G}_{c,p,-b,t_1} + \boldsymbol{\beta_B}\ SaleCounts^{B}_{c,p,-b,t_1} + \Theta' W_{i,c,p,b,t_0} + \lambda_c + \kappa_p + \psi_b + \varepsilon_{i,c,p,b,t_2}.$$

- **Price Discovery Mechanism: $\beta_G < 0$ and $\beta_B < 0$**
  Both types of events can reduce uncertainty and subsequent extension.

◂ Alternative channels
Appendix
◂ Additional Tests

Good and bad price news are both followed by lower extension rates

	(1) All sale payoffs	(2) + Mean price news	(3) Good vs. bad news
All peer sale payoffs	-0.668***	-0.656***	
	(0.140)	(0.138)	
Mean transaction price news		**-0.054**	
		(0.062)	
Good-news transactions (β_G)			**-0.671****
			(0.161)
Bad-news transactions (β_B)			**-0.691****
			(0.276)
Price-news cutoff	–	–	Pooled mean
Local controls	Y	Y	Y
Bank FE	Y	Y	Y
County FE, Property Type FE	Y	Y	Y
Market Size and Loan Level Controls	Y	Y	Y
Disclosure states only	Y	Y	Y
Observations	544,506	544,506	544,506
R-squared	0.0722	0.0722	0.0722

Prediction 2: Joint Specification

Include both sale payoffs and defaults.
Let $D_c = 1$ for counties in states requiring public disclosure of sale prices.

$$\begin{aligned} Ext_{i,c,p,b,t_2} = \alpha &+ \beta_S \, SaleCounts_{c,p,-b,t_1} + \beta_D \, DefaultCounts_{c,p,-b,t_1} \\ &+ \boldsymbol{\theta_S} \, (SaleCounts_{c,p,-b,t_1} \times D_c) \\ &+ \boldsymbol{\theta_D} \, (DefaultCounts_{c,p,-b,t_1} \times D_c) \\ &+ \Theta' W_{i,c,p,b,t_0} + \lambda_c + \kappa_p + \psi_b + \varepsilon_{i,c,p,b,t_2}. \end{aligned}$$

- **Sale payoffs** $\times D_c$**:** $\theta_S < 0$**.**
  A larger reduction in extension where prices are publicly disclosed.
- **Defaults** $\times D_c$**:** $\theta_D \approx 0$**.**
  No comparable disclosure difference expected.

Additional Tests

Prediction 2: Price disclosure strengthens the sale-payoff effect

Predictions Sale payoffs $\times$ D: $\theta_S < 0$; Defaults $\times$ D: $\theta_D \approx 0$

	(1) Baseline	(2) + Bank FE	(3) + Local controls
Sale payoffs $\times$ Disclosure (θ_S)	**-1.391****	**-0.812****	**-0.905****
	(0.601)	(0.385)	(0.407)
Defaults $\times$ Disclosure (θ_D)	5.402*	2.215	2.513
	(3.273)	(1.536)	(1.684)
Sale payoffs	0.330	-0.212	-0.092
	(0.554)	(0.319)	(0.337)
Defaults	-5.293	-2.065	-2.331
	(3.258)	(1.512)	(1.655)
Local controls	N	N	Y
Bank FE	N	Y	Y
County FE, Property Type FE	Y	Y	Y
Market Size and Loan Level Controls	Y	Y	Y
Observations	626,839	626,839	626,839
R-squared	0.0451	0.0728	0.0729

Notes: Standard errors clustered by county.

Control definitions

Prediction 2: Default-only Specification

Let $D_c = 1$ for counties in states requiring public disclosure of sale prices, and zero otherwise.

$$\begin{aligned} Ext_{i,c,p,b,t_2} = \alpha &+ \beta_D \, DefaultCounts_{c,p,-b,t_1} \\ &+ \boldsymbol{\theta_D} \left(DefaultCounts_{c,p,-b,t_1} \times D_c \right) \\ &+ \Theta' W_{i,c,p,b,t_0} + \lambda_c + \kappa_p + \psi_b + \varepsilon_{i,c,p,b,t_2}. \end{aligned}$$

- **Without required price disclosure ($D_c = 0$):**
  β_D measures the relationship between earlier defaults and subsequent extensions.
- **With required price disclosure ($D_c = 1$):**
  The corresponding relationship is $\beta_D + \theta_D$.

◂ Additional Tests

▸ Default-only results

Prediction 2: Default-only Results

	(1) Baseline	(2) + Bank FE	(3) + Local controls
Defaults × Disclosure (θ_D)	**2.767****	**1.252****	**1.179***
	(1.407)	(0.636)	(0.703)
Defaults	-3.311**	-1.693**	-1.568**
	(1.478)	(0.681)	(0.756)
WFH exposure	0.205	0.260**	0.354**
	(0.143)	(0.132)	(0.140)
Local controls	N	N	Y
Bank FE	N	Y	Y
County FE, Property Type FE	Y	Y	Y
Market Size and Loan Level Controls	Y	Y	Y
Observations	626,839	626,839	626,839
R-squared	0.0451	0.0728	0.0728

Notes: Standard errors clustered by county. Control definitions

Default-only specification

Appendix

Additional Tests

Robustness test on window cutoff

Quarter Cutoff	2021Q2 (1)	2021Q3 (2)	2021Q4 (3)	2022Q1 (4)	2022Q2 (5)
Coefficient on earlier resolutions at other banks					
Baseline OLS	-0.268	-0.241	-0.277^{**}	-0.246^{**}	-0.250^{***}
	(0.200)	(0.154)	(0.136)	(0.111)	(0.096)
Adding bank FE	-0.265^{**}	-0.248^{***}	-0.267^{***}	-0.232^{***}	-0.241^{***}
	(0.112)	(0.074)	(0.078)	(0.064)	(0.058)
Adding local controls and R	-0.234^{**}	-0.228^{***}	-0.256^{***}	-0.229^{***}	-0.229^{***}
	(0.116)	(0.075)	(0.078)	(0.065)	(0.055)
Mean extension rate (%)	3.79	3.56	3.31	3.05	2.78
Observations	674,188	651,724	626,839	606,900	585,321

◂ Additional Tests